

Supplementary Materials

for

“Computational Analysis of Ca^{2+} Dynamics in Isolated Cardiac Mitochondria Predicts Two Distinct Modes of Ca^{2+} Uptake”

by

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Appendix A: Model of Mitochondrial Oxidative Phosphorylation and Ca^{2+} Dynamics.

This appendix provides a brief description of the different reaction and transport fluxes in our integrated model of mitochondrial bioenergetics and Ca^{2+} dynamics (see Figure 1A of the main document). The key components of the present integrated model are based on our previously developed model of oxidative phosphorylation (OxPhos) and Ca^{2+} dynamics (1), which was developed from the original model of OxPhos and Ca^{2+} dynamics (2, 3) to better characterize the mitochondrial Ca^{2+} uptake, egress and sequestration mechanisms based on the experimental data of Blomeyer et al. (4) in isolated guinea pig cardiac mitochondria. In summary, the previous model (1) included the kinetics of the reactions at complexes I, III, IV and V (F_1F_0 -ATPase) of the respiratory system, metabolite transporters ANT (adenine nucleotide translocase) and PHT (phosphate/ H^+ cotransporter or phosphate/ OH^- exchanger), and cation transport fluxes across the IMM, including the fluxes via the KHE (K^+/H^+ exchanger), NHE (Na^+/H^+ exchanger), CHE ($\text{Ca}^{2+}/\text{H}^+$ exchanger), NCE ($\text{Na}^+/\text{Ca}^{2+}$ exchanger), CU (Ca^{2+} uniporter), RaM (rapid-mode of Ca^{2+} uptake), and passive H^+ permeation (“leak”). One of the key features of the present integrated model is the addition of a putative mitochondrial RTC (RyR-type channel) that characterizes the fast Ca^{2+} uptake model observed in the experiments of Boelens et al. (5) in isolated guinea pig cardiac mitochondria, which is distinct from the slow Ca^{2+} uptake mode via the CU. Altogether, the present integrated model has a strong potential to provide very accurate simulation of mitochondrial $[\text{Ca}^{2+}]$ dynamics under physiological variations in cytosolic cations. In addition, the present integrated model considers the binding of free cations (H^+ , K^+ , Mg^{2+} , Ca^{2+} and Na^+) to various metabolites present in both extra-matrix (e.g., ATP, ADP, Pi and EGTA) and matrix (e.g., ATP, ADP, Pi and other buffering proteins) compartments of isolated mitochondrial system. For simplicity, these cation-metabolite bindings are assumed to occur very fast so that the reactions are always at near equilibrium. These simplifications reduce the number of differential equations to solve that govern the biochemical processes of energy metabolism and cation handling in mitochondria. The governing model equations and the associated model parameters along with their previously estimated values (see Table 1) are summarized below.

Respiratory System and Oxidative Phosphorylation:

Dehydrogenase

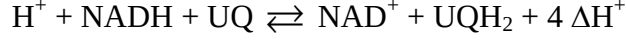


$$J_{\text{DH}} = X_{\text{DH}} \left(\frac{1 + [\text{Pi}]_x / k_{\text{Pi},1}}{1 + [\text{Pi}]_x / k_{\text{Pi},2}} \right) (r_{\text{DH}} [\text{NAD}^+]_x - [\text{NADH}]_x) \quad (\text{A.1})$$

The kinetics of TCA cycle and other NADH-producing reactions were not taken into account in the present integrated model. Here the rate of NADH production is modeled using a phosphate-

dependent phenomenological dehydrogenase (DH) flux expression. Note that, the subscripts ‘x’, ‘i’ and ‘e’ are used to represent the concentrations of species/ions with respect to matrix, inner membrane and extra-matrix spaces.

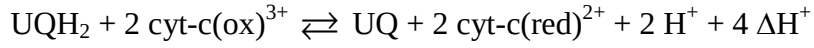
Complex I



$$J_{\text{C1}} = X_{\text{C1}} \left(\frac{\exp\left(-\left(\Delta G_{\text{o,C1}} + 4\Delta G_{\text{H}} - RT \ln([\text{H}^+]_{\text{x}} / 10^{-7})\right)\right)/RT}{\times [\text{NADH}]_{\text{x}} [\text{UQ}] - [\text{NAD}^+]_{\text{x}} [\text{UQH}_2]} \right) \quad (\text{A.2})$$

where $\Delta G_{\text{H}} = F\Delta\Psi + RT \ln([\text{H}^+]_{\text{e}} / [\text{H}^+]_{\text{x}})$ is the change in Gibb’s free energy associated with the pumping of protons from the matrix side to the extra-matrix side of the IMM. Each time this reaction proceeds, a total of 4 protons are pumped out from the matrix to cytosol.

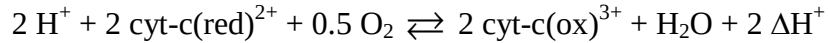
Complex III



$$J_{\text{C3}} = X_{\text{C3}} \left(\frac{1 + [\text{Pi}]_{\text{x}} / k_{\text{Pi,3}}}{1 + [\text{Pi}]_{\text{x}} / k_{\text{Pi,4}}} \right) \left(\frac{\exp\left(-\left(\Delta G_{\text{o,C3}} + 4\Delta G_{\text{H}} + 2RT \ln([\text{H}^+]_{\text{x}} / 10^{-7}) - 2F\Delta\Psi\right)\right)/2RT}{\times [\text{cyt-c(ox)}^{3+}] [\text{UQH}_2]^{1/2} - [\text{cyt-c(red)}^{2+}] [\text{UQ}]^{1/2}} \right) \quad (\text{A.3})$$

The model accounts for a phosphate-dependent control of the complex III reaction. Each time this reaction proceeds, a total of 2 protons are pumped out from the matrix to cytosol.

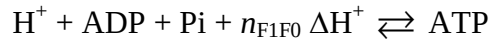
Complex IV



$$J_{\text{C4}} = X_{\text{C4}} \left(\frac{[\text{O}_2]}{k_{\text{O}_2} + [\text{O}_2]} \right) \left(\frac{[\text{cyt-c(red)}^{2+}]}{\text{cyt-c}_{\text{tot}}} \right) \left(\frac{\exp\left(-\left(\Delta G_{\text{o,C4}} + 2\Delta G_{\text{H}} - 2RT \ln([\text{H}^+]_{\text{x}} / 10^{-7})\right)\right)/2RT}{\times [\text{cyt-c(red)}^{2+}] [\text{O}_2]^{1/4} - e^{+F\Delta\Psi/RT} [\text{cyt-c(ox)}^{3+}]} \right) \quad (\text{A.4})$$

Each time this reaction proceeds (oxygen consumption), a total of 2 protons are pumped from the matrix to cytosol.

Complex V (F_1F_0 -ATPase)



$$J_{\text{F1F0}} = X_{\text{F1F0}} \left(\frac{\exp\left(-\left(\Delta G_{\text{o,F1F0}} - n_{\text{F1F0}} \Delta G_{\text{H}} - RT \ln([\text{H}^+]_{\text{x}} / 10^{-7})\right) / RT\right)}{\times \frac{[\text{P}_{\text{ATP}}]_{\text{x}}}{[\text{P}_{\text{ADP}}]_{\text{x}} [\text{P}_{\text{Pi}}]_{\text{x}}} \times [\text{ADP}]_{\text{x}} [\text{Pi}]_{\text{x}} - [\text{ATP}]_{\text{x}}} \right) \quad (\text{A.5})$$

Each time this reaction proceeds (ATP synthesis), a total of n_{F1F0} ($= 3$) protons enter into the matrix to cytosol, through the proton channel of F_1F_0 -ATPase.

Adenine Nucleotide Translocase (ANT)

$$J_{\text{ANT}} = X_{\text{ANT}} \left(\frac{[\text{fADP}]_{\text{i}}}{k_{\text{ADP}} + [\text{fADP}]_{\text{i}}} \right) \left(\frac{[\text{fADP}]_{\text{i}}}{[\text{fADP}]_{\text{i}} + [\text{fATP}]_{\text{i}} e^{-0.35 F \Delta \Psi / RT}} - \frac{[\text{fADP}]_{\text{x}}}{[\text{fADP}]_{\text{x}} + [\text{fATP}]_{\text{x}} e^{+0.65 F \Delta \Psi / RT}} \right) \quad (\text{A.6})$$

The exchange of ATP for ADP via the ANT is electrogenic, so the flux is dependent on the IMM potential $\Delta \Psi$. In addition, the exchange is regulated (activated) by the extra-matrix free ADP concentration ($[\text{ADP}]_{\text{e}} \approx ([\text{ADP}]_{\text{i}})$).

Phosphate-Hydrogen Cotransporter (PHT)

$$J_{\text{PHT}} = X_{\text{PHT}} \left(\frac{[\text{H}_2\text{PO}_4^-]_{\text{x}} [\text{H}^+]_{\text{x}} - [\text{H}_2\text{PO}_4^-]_{\text{i}} [\text{H}^+]_{\text{i}}}{[\text{H}_2\text{PO}_4^-]_{\text{i}} + k_{\text{PHT}}} \right) \quad (\text{A.7})$$

where $[\text{H}_2\text{PO}_4^-]_{\text{i}} = \frac{[\text{H}^+]_{\text{i}} [\text{Pi}]_{\text{i}}}{([\text{H}^+]_{\text{i}} + k_{\text{dH}})}$, $[\text{H}_2\text{PO}_4^-]_{\text{x}} = \frac{[\text{H}^+]_{\text{x}} [\text{Pi}]_{\text{x}}}{([\text{H}^+]_{\text{x}} + k_{\text{dH}})}$ and $[\text{H}^+]_{\text{i}} = [\text{H}^+]_{\text{e}}$.

Potassium-Hydrogen Exchanger (KHE)

$$J_{\text{KHE}} = X_{\text{KHE}} \left([\text{K}^+]_{\text{x}} [\text{H}^+]_{\text{i}} - [\text{K}^+]_{\text{i}} [\text{H}^+]_{\text{x}} \right); [\text{K}^+]_{\text{i}} = [\text{K}^+]_{\text{e}} \text{ and } [\text{H}^+]_{\text{i}} = [\text{H}^+]_{\text{e}} \quad (\text{A.8})$$

Passive Proton Permeation (Leak)

$$J_{\text{Hleak}} = X_{\text{Hleak}} \Delta \Psi \left(\frac{[\text{H}^+]_{\text{i}} \exp(+F \Delta \Psi / RT) - [\text{H}^+]_{\text{x}}}{\exp(+F \Delta \Psi / RT) - 1} \right); [\text{H}^+]_{\text{i}} = [\text{H}^+]_{\text{e}} \quad (\text{A.9})$$

Passive Substrate Transport Fluxes (Outer Mitochondrial Membrane)

$$\begin{aligned} J_{\text{ATPt}} &= \gamma_{\text{m}} p_{\text{A}} ([\text{ATP}]_{\text{e}} - [\text{ATP}]_{\text{i}}) \\ J_{\text{ADPt}} &= \gamma_{\text{m}} p_{\text{A}} ([\text{ADP}]_{\text{e}} - [\text{ADP}]_{\text{i}}) \\ J_{\text{Pi}} &= \gamma_{\text{m}} p_{\text{Pi}} ([\text{Pi}]_{\text{e}} - [\text{Pi}]_{\text{i}}) \end{aligned} \quad (\text{A.10})$$

Table A.1: Fixed parameter values in the model of mitochondrial respiratory system and oxidative phosphorylation. Most of the oxidative phosphorylation model parameter values are from Dash and Beard (3) in isolated cardiac mitochondria (“mito” denotes mitochondria) and the details can be found in our earlier works on mitochondrial bioenergetics (2, 6-9). Here, r1 refers to standard physicochemical/thermodynamic constants.

Parameter	Description	Values	Units	Ref.
<i>Physicochemical/Thermodynamic Parameters:</i>				
RT	Gas constant times temperature	2.4788	kJ mol^{-1}	-r1
F	Faraday’s constant	0.096484	$\text{kJ mol}^{-1} \text{mV}^{-1}$	-r1
$\Delta G_{o,C1}$	Standard free energy, complex I	-69.37	kJ mol^{-1}	(1-3, 10)
$\Delta G_{o,C3}$	Standard free energy, complex III	-32.53	kJ mol^{-1}	(1-3, 10)
$\Delta G_{o,C4}$	Standard free energy, complex IV	-122.94	kJ mol^{-1}	(1-3, 10)
$\Delta G_{o,F1F0}$	Standard free energy, F_1F_0 -ATPase	36.03	kJ mol^{-1}	(1-3, 10)
<i>Anatomical Parameters:</i>				
W_x	Matrix water fraction	0.6435	$(\text{l water}) (\text{l mito})^{-1}$	(1-3, 11)
W_i	IMS water fraction	0.0715	$(\text{l water}) (\text{l mito})^{-1}$	(1-3, 11)
W_e	External buffer water fraction	2000 [†]	$(\text{l water}) (\text{l mito})^{-1}$	(1, 4, 5)
γ_m	Outer membrane area per mito volume	5.99	μm^{-1}	(1-3, 12)
<i>Fixed Concentrations Pools:</i>				
NAD_{tot}	Total matrix NAD concentration	3.0	$\text{mmol} (\text{l matrix water})^{-1}$	(1-3, 12)
UQ_{tot}	Total matrix ubiquinol concentration	1.35	$\text{mmol} (\text{l matrix water})^{-1}$	(1-3, 12)
$\text{cyt-c}_{\text{tot}}$	Total IMS cytochrome-c concentration	2.70	$\text{mmol} (\text{l IMS water})^{-1}$	(1-3, 12)
A_{tot}	Total matrix adenine nucleotide concentration (ATP+ADP)	10.0	$\text{mmol} (\text{l matrix water})^{-1}$	(1-3, 12)
$[\text{O}_2]$	O_2 concentration (constant)	3.48×10^{-5}	M	(1, 4, 5)
<i>Oxidative Phosphorylation Model Parameters:</i>				
$k_{\text{Pi},1}$	Dehydrogenase/Pi parameter	0.1553	mM	(1-3)
$k_{\text{Pi},2}$	Dehydrogenase/Pi parameter	0.8222	mM	(1-3)
r_{DH}	Dehydrogenase phenomenological parameter	4.559	unitless	(1-3)
X_{DH}	Dehydrogenase activity	0.0866	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	(1-3)
X_{C1}	Complex I activity	4.405×10^3	$\text{mol s}^{-1} \text{M}^{-2} (\text{l mito})^{-1}$	(1-3)
X_{C3}	Complex III activity	4.887	$\text{mol s}^{-1} \text{M}^{-3/2} (\text{l mito})^{-1}$	(1-3)
X_{C4}	Complex IV activity	6.766×10^{-5}	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	(1-3)
X_{F1F0}	F_1F_0 -ATPase activity	1000.0	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	(1-3)
X_{ANT}	ANT activity	8.123×10^{-3}	$\text{mol s}^{-1} (\text{l mito})^{-1}$	(1-3)
n_{F1F0}	H^+ stoichiometry for F_1F_0 -ATPase	3.0	unitless	(1-3)
X_{PHT}	Pi^-/H^+ cotransporter activity	3.850×10^5	$\text{mol s}^{-1} \text{M}^{-1} (\text{l mito})^{-1}$	(1-3)

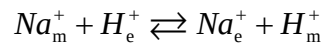
k_{PHT}	Pi^-/H^+ cotransporter parameter	0.2542	mM	(1-3)
X_{KHE}	K^+/H^+ antiporter activity	5.651×10^7	$\text{mol s}^{-1} \text{M}^{-2} (\text{l mito})^{-1}$	(1-3)
X_{Hleak}	Proton leak activity	200.0	$\text{mol s}^{-1} \text{M}^{-1} \text{mV}^{-1} (\text{l mito})^{-1}$	(1-3)
$k_{\text{Pi},3}$	Complex III/Pi parameter	0.3601	mM	(1-3)
$k_{\text{Pi},4}$	Complex III/Pi parameter	5.924	mM	(1-3)
p_{Pi}	Outer mitochondrial membrane permeability to inorganic phosphate	327.0	$\mu\text{m sec}^{-1} (\text{l mito})^{-1}$	(1-3)
p_{A}	Outer mitochondrial membrane permeability to nucleotides	85.0	$\mu\text{m sec}^{-1} (\text{l mito})^{-1}$	(1-3)
k_{ADP}	ANT Michaelis-Menten constant	3.5×10^{-6}	M	(1-3, 12)
k_{O_2}	Kinetic constant for complex IV	1.2×10^{-4}	M	(1-3, 12)
C_{IMM}	Effective capacitance of IMM	6.75×10^{-6}	$\text{mol (l mito)}^{-1} \text{mV}^{-1}$	(1-3)
<i>Buffer/IMS ion concentrations</i>				
$[\text{Ca}^{2+}]_{\text{e}, \text{i}}$	Buffer/IMS Ca^{2+} ion concentration	Variable	$\text{mmol (l buffer water)}^{-1}$	(4, 5)
$[\text{Na}^+]_{\text{e}, \text{i}}$	Buffer/IMS Na^+ ion concentration	Variable	$\text{mmol (l buffer water)}^{-1}$	(4, 5)
$[\text{Mg}^{2+}]_{\text{e}, \text{i}}$	Buffer/IMS Mg^{2+} ion concentration	Variable	$\text{mmol (l buffer water)}^{-1}$	(4, 5)
$[\text{H}^+]_{\text{e}, \text{i}}$	Buffer/IMS H^+ ion concentration	$10^{-7.15}$	$\text{mmol (l buffer water)}^{-1}$	(4, 5)
$[\text{K}^+]_{\text{e}, \text{i}}$	Buffer/IMS K^+ ion concentration	130	$\text{mmol (l buffer water)}^{-1}$	(4, 5)
$[\text{Pi}]_{\text{e}, \text{i}}$	Buffer/IMS Pi concentration	5	$\text{mmol (l buffer water)}^{-1}$	(4, 5)

† 0.5 mg mito protein in the experimental buffer with 1 mg mito protein = 1 μL of water space.

Sodium and Calcium Transport Fluxes:

Sodium Hydrogen Exchanger (NHE)

The biochemical reaction for the NHE is defined as:



The rate expression used in the model is:

$$J_{\text{NHE}} = \frac{X_{\text{NHE}} [\text{H}^+]_m^3}{(K_{\text{IH}}^3 + [\text{H}^+]_m^3)} \times \left(\frac{[\text{H}^+]_e [\text{Na}^+]_m - [\text{H}^+]_m [\text{Na}^+]_e}{K_{\text{Na}} ([\text{H}^+]_e + [\text{H}^+]_m) + [\text{H}^+]_e^2 [\text{Na}^+]_m + [\text{H}^+]_m [\text{Na}^+]_e} \right) \quad (\text{A.11})$$

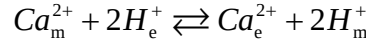
The factor $\frac{[\text{H}^+]_m^3}{(K_{\text{IH}}^3 + [\text{H}^+]_m^3)}$ represents the inhibition of the NHE by alkaline matrix pH ($\text{pH}_m > 7$).

Table A.2: Sodium-Hydrogen Exchanger Parameters

Parameter	Definition	Value	Unit	Ref.
X_{NHE}	NHE activity	14×10^{-9}	$\text{mol sec}^{-1} \text{mg protein}^{-1}$	(1)
K_{iH}	H^+ inactivation constant	$10^{-7.2}$	M	(1)
K_{Na}	Na^+ binding constant	25	mM	(1)

Calcium Hydrogen Exchanger (CHE)

The biochemical reaction for the CHE is defined as:



The rate expression used in the model is:

$$J_{CHE} = X_{CHE} \left(\frac{[H^+]_e^2 [Ca^{2+}]_m - [H^+]_m^2 [Ca^{2+}]_e}{\left(K_{Ca,CHE} ([H^+]_e^2 + [H^+]_m^2) + [H^+]_e^2 [Ca^{2+}]_m + [H^+]_m^2 [Ca^{2+}]_e \right)} \right) \quad (A.12)$$

The exchange of Ca^+ for H^+ via the CHE is assumed to be electroneutral ($n_{CHE} = 2$), and hence the flux is independent of IMM potential $\Delta\Psi$.

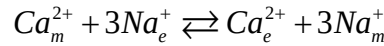
Table A.3: Calcium-Hydrogen Exchanger Parameters

Parameter	Definition	Value	Unit	Ref.
X_{CHE}	CHE activity	0.94×10^{-9}	$\text{mol sec}^{-1} \text{mg protein}^{-1}$	(13) [†]
$K_{Ca,CHE}$	Ca^{2+} binding constant	4.8×10^{-6}	M	(13)

[†] Adjusted to obtain better fits to the data of Boelens et al. (5). While simulating the data-set of Blomeyer et al. (4), this value was set to 2×10^{-9} .

Sodium Calcium Exchanger (NCE)

The biochemical reaction for the NCE is defined as:



The rate expression used in the model is:

$$J_{NCE} = \frac{X_{NCE}}{D} \left(k_a \cdot D_{Hm} \frac{[Ca^{2+}]_m [Na^+]_e^n}{K'_{Cx} \cdot K_{Na}^n} - k_b \cdot D_{He} \frac{[Ca^{2+}]_e [Na^+]_m^n}{K'_{Ce} \cdot K_{Na}^n} \right) \quad (A.13)$$

where the binding constants are:

$$K'_{\text{Cm}} = K_{\text{Ca}} \left(\sum_{q=1}^m \frac{[\text{H}^+]_m^q}{K_{\text{H1}}^q} + 1 + \sum_{q=1}^m \frac{K_{\text{H2}}^q}{[\text{H}^+]_m^q} \right)$$

$$K'_{\text{Ce}} = K_{\text{Ca}} \left(\sum_{q=1}^m \frac{[\text{H}^+]_e^q}{K_{\text{H1}}^q} + 1 + \sum_{q=1}^m \frac{K_{\text{H2}}^q}{[\text{H}^+]_e^q} \right)$$

and the $\Delta\Psi$ -dependent translocation terms are:

$$k_a = e^{[(n\beta z_{\text{Na}} - \beta z_{\text{Ca}})\Delta\Phi]}$$

$$k_b = e^{[(-n\beta z_{\text{Na}} + \beta z_{\text{Ca}})\Delta\Phi]}$$

with the denominator terms defined as:

$$D_{\text{Hm}} = \left(\sum_{q=1}^m \frac{[\text{H}^+]_m^q}{K_{\text{H1}}^q} + 1 + \sum_{q=1}^m \frac{K_{\text{H2}}^q}{[\text{H}^+]_m^q} \right) \left(\frac{[\text{H}^+]_m^m}{K_{\text{H2}}^m} \right)$$

$$D_{\text{He}} = \left(\sum_{q=1}^m \frac{[\text{H}^+]_e^q}{K_{\text{H1}}^q} + 1 + \sum_{q=1}^m \frac{K_{\text{H2}}^q}{[\text{H}^+]_e^q} \right) \left(\frac{[\text{H}^+]_e^m}{K_{\text{H2}}^m} \right)$$

$$D = D_{\text{Hm}} \left(1 + \sum_{p=1}^n \frac{[\text{Na}^+]_e^p}{K'_{\text{Ne}}{}^p} + \frac{[\text{Ca}^{2+}]_m}{K'_{\text{Cm}}} + \sum_{p=1}^n \frac{[\text{Ca}^{2+}]_m [\text{Na}^+]_e^p}{K'_{\text{Cm}} \cdot K'_{\text{Ne}}{}^p} \right) + D_{\text{He}} \left(1 + \sum_{p=1}^n \frac{[\text{Na}^+]_m^p}{K'_{\text{Nm}}{}^p} + \frac{[\text{Ca}^{2+}]_e}{K'_{\text{Ce}}} + \sum_{p=1}^n \frac{[\text{Ca}^{2+}]_e [\text{Na}^+]_m^p}{K'_{\text{Ce}} \cdot K'_{\text{Nm}}{}^p} \right) - 1$$

The $3\text{Na}^+ - 1\text{Ca}^{2+}$ antiporter model (Model 2 Case 1) of Pradhan et al (14) is used to describe the Na^+ dependent Ca^{2+} extrusion. This model depicts the experimentally observed pH inhibition of the antiporter function. The flux is dependent on the IMM potential $\Delta\Psi$ due to the electrogenic nature of the NCE.

Table A.4: Sodium-Calcium Exchanger Parameters

Parameter	Definition	Value	Unit	Ref.
X_{NCE}	$\text{Na}^+/\text{Ca}^{2+}$ antiporter activity	7.84×10^{-3}	$\text{mol sec}^{-1} \text{mg protein}^{-1}$	(14) [†]
K_{Na}	True dissociation constant of Na^+ for the antiporter	2.2×10^{-3}	M	(14)
K_{Ca}	True dissociation constant of Ca^{2+} for the antiporter	1.1×10^{-7}	M	(1)
K_{H1}	1 st Proton binding constant	6.4×10^{-8}	M	(1, 14)
K_{H2}	2 nd Proton binding constant	1.4×10^{-7}	M	(1, 14)
$\alpha_e = \alpha_x = \alpha$	Biophysical parameters of NCE model	0.0	Unitless	(14)
$\beta_e = \beta_x = \beta$	Biophysical parameters of NCE model	0.5	Unitless	(14)
n	NCE stoichiometry parameter	3.0	Unitless	(14)
m	Proton regulatory parameter	3.0	Unitless	(1)

[†] Adjusted to obtain better fits to the data of Blomeyer et al. (4).

Calcium Uniporter (CU)

This is the simplified version of the CU model by Pradhan et al. (15) which depicts the experimentally observed Mg^{2+} inhibition of mitochondrial Ca^{2+} uptake via the CU and Pi regulation of the CU function at submillimolar levels of cytosolic Pi. Note that this simplified model accounts for inhibition of the CU flux only by cytosolic Mg^{2+} , based on recent structural data on the CU (16, 17). Revised schematic of the CU is shown in Figure 1C of the main document. Three kinetic models (full cooperativity; partial cooperativity; no cooperativity) are simulated to exclusively fit the available experimental data sets (18-24) on the kinetics of extra-matrix Ca^{2+} and $\Delta\Psi$ dependent mitochondrial Ca^{2+} uptake via the CU measured both in the absence and presence of different amounts of Mg^{2+} and Pi in the experimental buffer. Model fittings of the best model, which assumes partial cooperativity of Ca^{2+} and Mg^{2+} binding with the CU, for these data sets is shown in Figure A.1. The model equations describing the Ca^{2+} flux via the CU are given by the following equations:

$$J_{\text{CU}} = \frac{[\text{T}]_{\text{tot}}}{D} \left(k_i \frac{[\text{Ca}^{2+}]_e^2}{K_{\text{Ce}}^2} - k_o \frac{[\text{Ca}^{2+}]_x^2}{K_{\text{Cx}}^2} \right),$$

$$K_{\text{Ce}} = K_{\text{Ce},1} \exp(-\alpha_e \Delta\Phi), \quad K_{\text{Ce},1} = K_{\text{Ce}}^0 \cdot (1 + [\text{Pi}]_e / (K_{\text{Pi}} + [\text{Pi}]_e));$$

$$K_{\text{Cx}} = K_{\text{Cx},1} \exp(+\alpha_x \Delta\Phi), \quad K_{\text{Cx},1} = K_{\text{Cx}}^0 \cdot (1 + [\text{Pi}]_x / (K_{\text{Pi}} + [\text{Pi}]_x));$$

$$K_{\text{Me}} = K_{\text{Me},1} \exp(-\alpha_e \Delta\Phi), \quad K_{\text{Me},1} = K_{\text{Me}}^0 / (1 + [\text{Pi}]_e / (K_{\text{Pi}} + [\text{Pi}]_e));$$
(A.14)

$$k_i = k_i^0 \exp(+2\beta_e \Delta\Phi);$$

$$\beta_e = \frac{1}{2} \left[1 + \frac{nH}{\Delta\Phi} \ln \left\{ (\Delta\Phi / nH) / \sinh(\Delta\Phi / nH) \right\} \right] - \alpha_e,$$

$$k_o = k_o^0 \exp(-2\beta_x \Delta\Phi); \quad k_o^0 = k_i^0 (K_{\text{Cx},1} / K_{\text{Ce},1})^2,$$

$$\beta_x = \frac{1}{2} \left[1 - \frac{nH}{\Delta\Phi} \ln \left\{ (\Delta\Phi / nH) / \sinh(\Delta\Phi / nH) \right\} \right] - \alpha_x; \quad \Delta\Phi = F \Delta\Psi / RT.$$
(A.15)

$$D = 1 + \frac{[\text{Ca}^{2+}]_e}{K_{\text{Ce}}} + \frac{[\text{Ca}^{2+}]_e^2}{K_{\text{Ce}}^2} + \frac{[\text{Ca}^{2+}]_x}{K_{\text{Cx}}} + \frac{[\text{Ca}^{2+}]_x^2}{K_{\text{Cx}}^2} + \frac{[\text{Mg}^{2+}]_e}{K_{\text{Me}}} + \frac{[\text{Mg}^{2+}]_e^2}{K_{\text{Me}}^2}$$

$$+ \frac{[\text{Ca}^{2+}]_e [\text{Mg}^{2+}]_e}{\gamma \cdot K_{\text{Ce}} \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_e^2 [\text{Mg}^{2+}]_e^2}{\gamma^2 \cdot K_{\text{Ce}} \cdot K_{\text{Me}}^2} + \frac{[\text{Ca}^{2+}]_e^2 [\text{Mg}^{2+}]_e}{\gamma^2 \cdot K_{\text{Ce}}^2 \cdot K_{\text{Me}}} + \frac{[\text{Ca}^{2+}]_e^2 [\text{Mg}^{2+}]_e^2}{\gamma^4 \cdot K_{\text{Ce}}^2 \cdot K_{\text{Me}}^2}.$$
(A.16)

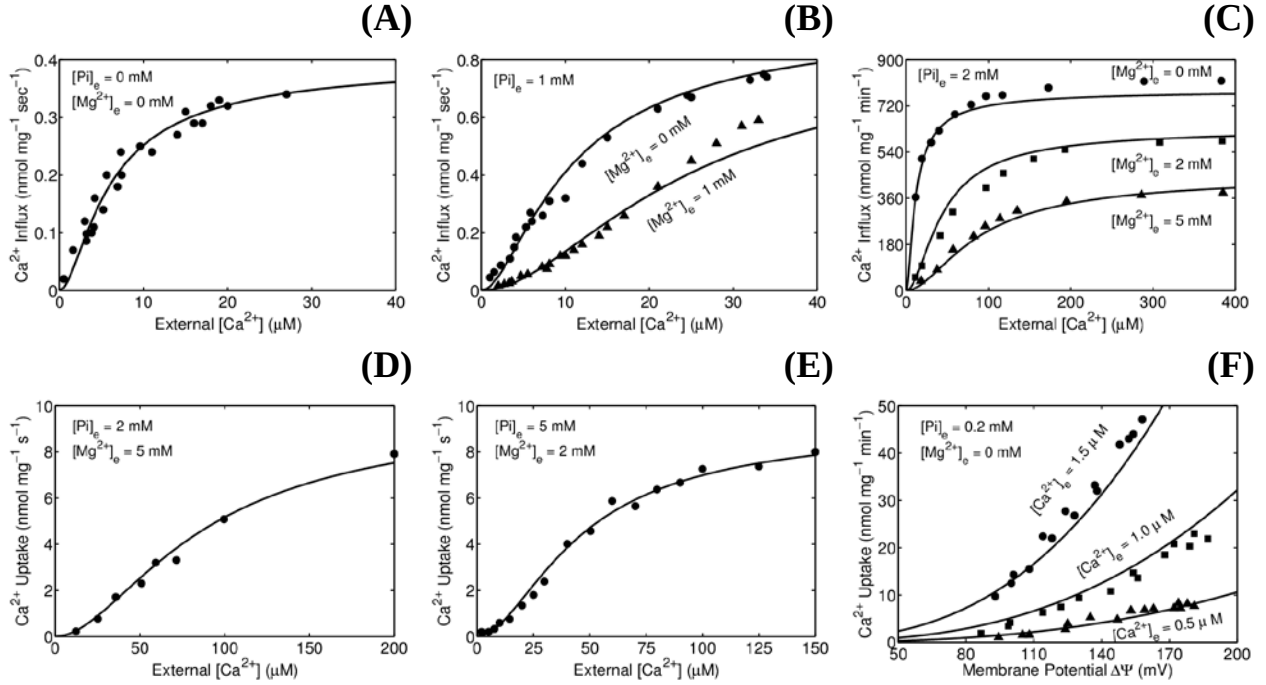


Figure A.1: Comparison of the Ca^{2+} uniporter models (lines) to the experimental data (points) on the extra-matrix Ca^{2+} and $\Delta\Psi$ dependencies of mitochondrial Ca^{2+} uptake measured both in the presence and absence of extra-matrix Mg^{2+} . Shown are the fittings of the best versions of the kinetic model (partial cooperativity) to the kinetic data of (A,B) Crompton et al. (25) in which the initial rates of Ca^{2+} influx in response to variations in extra-matrix $[\text{Ca}^{2+}]$ were measured in energized mitochondria of rat hearts suspended (A) in a medium containing 0 mM Mg^{2+} and 0 mM Pi and (B) in a medium containing 0 and 1 mM Mg^{2+} and 1 mM Pi , (C) Bragadin et al. (19) in which the initial rates of Ca^{2+} influx in response to variations in extra-matrix $[\text{Ca}^{2+}]$ were measured in energized mitochondria of rat livers suspended in a medium containing different levels of Mg^{2+} ($[\text{Mg}^{2+}]_e = 0, 2$ and 5 mM) and 2 mM Pi , (D) Vinogradov and Scarpa (21) in which the initial rates of Ca^{2+} influx in response to variations in extra-matrix $[\text{Ca}^{2+}]$ were measured in energized mitochondria of rat livers suspended in a medium containing 2 mM Mg^{2+} and 5 mM Pi , (E) Scarpa and Graziotti (26) in which the initial rates of Ca^{2+} influx in response to variations in extra-matrix $[\text{Ca}^{2+}]$ were measured in energized mitochondria of rat hearts suspended in a medium containing 5 mM Mg^{2+} and 2 mM Pi , and (F) Wingrove et al. (23) in which the initial rates of Ca^{2+} influx as a function of $\Delta\Psi$ were measured in energized mitochondria of rat livers suspended in a medium containing three different levels of Ca^{2+} ($[\text{Ca}^{2+}]_e = 0.5, 1.0$ and $1.5 \mu\text{M}$), with 0 mM Mg^{2+} and 0.2 mM Pi present in the medium. The models were fitted to these data sets by setting $\Delta\Psi$ at 140 mV for the data corresponding to 0 mM Pi (A) and 190 mV for the data corresponding to $\geq 1 \text{ mM}$ Pi (B-E). Matrix $[\text{Ca}^{2+}]$ was fixed at 250 nM and matrix $[\text{Mg}^{2+}]$ was fixed at 0.4 mM.

Table A.5: Calcium Uniporter Parameters[†]

Parameter	Definition	Value	Unit	Ref.
K_C^0	True dissociation constant of extra-matrix and or matrix Ca^{2+} for uniporter	4.15×10^{-6}	M	-o
K_M^0	True dissociation constant of extra-matrix for uniporter	0.8×10^{-3}	M	-o
K_{Pi}	True dissociation constant of extra-matrix and or matrix Pi for uniporter	0.2×10^{-3}	M	(15)
γ	Control parameter in CU model	4.8	Unitless	-o
$\alpha_e = \alpha_x = \alpha$	Biophysical parameter of CU model	0.0	Unitless	(15)
nH	Biophysical parameters of CU model	2.65	Unitless	(15)

-o: optimized to fit the data from (18-24).

†: k_i^0 is the calcium uniporter activity, the value(s) is/are listed in Table 1 of the main document.

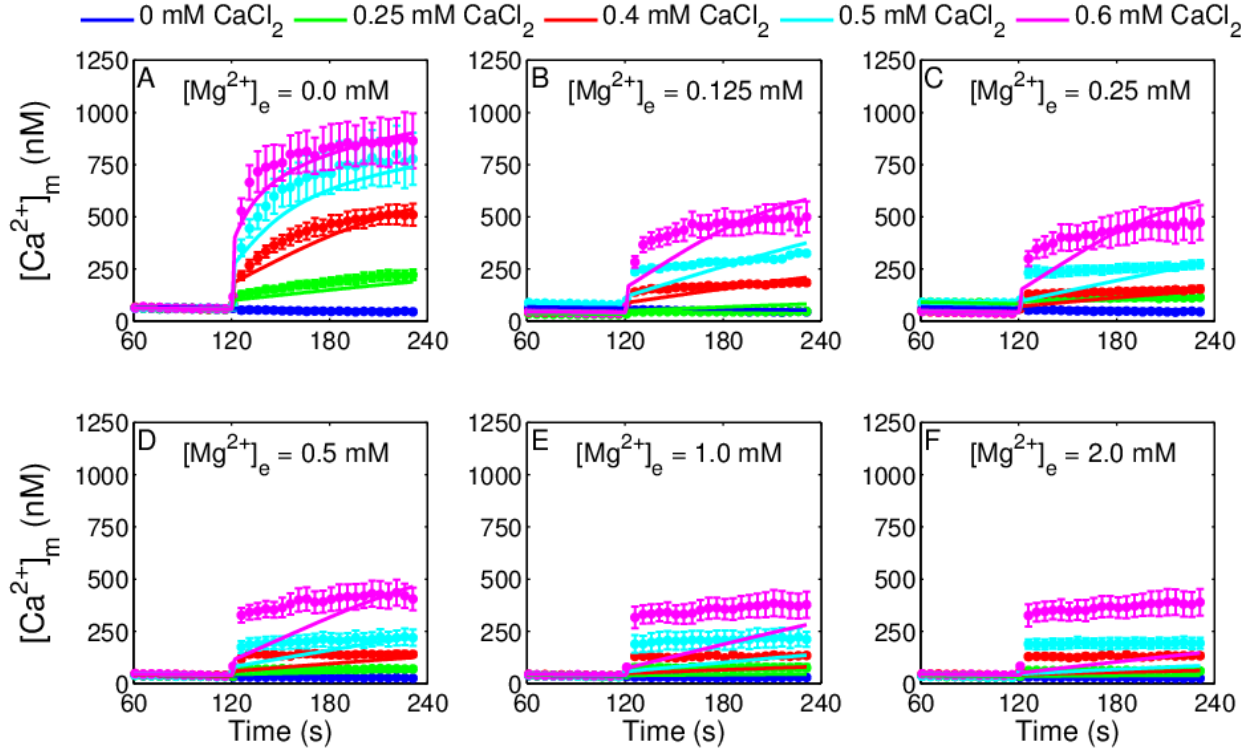


Figure A.2: Model simulations of the Ca^{2+} uptake experiments in the absence of RTC. Experimentally measured matrix $[\text{Ca}^{2+}]$ dynamics (closed circles with error bars) and model simulations (solid lines) in response to added CaCl_2 boluses of 0 mM (blue), 0.25 mM (green), 0.4 mM (red), 0.5 mM (cyan) and 0.6 mM (magenta). The experimental buffer contained either 0 mM (A), 0.125 mM (B), 0.25 mM (C), 0.5 mM (D), 1 mM (E), or 2 mM (F) MgCl_2 , also containing 1 mM of EGTA. The estimated parameters corresponding to the mitochondrial Ca^{2+} sequestration system and CU that provide these model simulations are listed in Table 1 of the main document.

RyR Type Channel (RTC)

The base model governing the gating kinetics of RTC is essentially the same (except for the Mg^{2+} binding step shown in Figure 1D of main document) as the Keizer-Levine model (27). However, for the sake of completeness here we provide the associated model equations for the schematic shown in Figure 1D of the main document:

$$\begin{aligned}
 \frac{dP_{C_1}}{dt} &= -k_a^+ [\text{Ca}^{2+}]_e^4 P_{C_1} + k_a^- P_{O_1}, \\
 \frac{dP_{O_1}}{dt} &= k_a^+ [\text{Ca}^{2+}]_e^4 P_{C_1} - k_a^- P_{O_1} - k_b^+ [\text{Ca}^{2+}]_e^3 P_{O_1} \\
 &\quad + k_b^- P_{O_2} - k_c^+ P_{O_1} + f(\text{Mg}^{2+}) k_c^- P_{C_2}, \\
 \frac{dP_{O_2}}{dt} &= k_b^+ [\text{Ca}^{2+}]_e^3 P_{O_1} - k_b^- P_{O_2}, \\
 \frac{dP_{C_2}}{dt} &= k_c^+ P_{O_1} - k_c^- P_{C_2}.
 \end{aligned} \tag{A.17}$$

Using the condition: $P_{C_1} + P_{C_2} + P_{O_1} + P_{O_2} = 1$, we eliminated one of the equations to yield three independent equations before integrating them with the mitochondrial model shown in Fig. 1A of the main document. $f(\text{Mg}^{2+})$ is the function given by equation (1) of the main document. It accounts for the Mg^{2+} dependent binding step shown in Figure 1D of the main document. Concentration dependent effect of Mg^{2+} binding with the C_2 state (as shown in Figure 1D of the main document) and O_1 state (the alternate model exhibiting rapid adaptation of RTC) is shown in Figure A.3 (E-H) and (I-L), respectively.

Table A.6: RyR-Type Channel Parameters (27)

Parameter	Definition	Value	Unit
k_a^+	Ca^{2+} -dependent rate constant from $C_1 \rightarrow O_1$	1500	$\mu\text{M}^{-4}\text{s}^{-1}$
k_a^-	Ca^{2+} -independent rate constant from $O_1 \rightarrow C_1$	28.8	s^{-1}
k_b^+	Ca^{2+} -dependent rate constant from $O_1 \rightarrow O_2$	1500	$\mu\text{M}^{-3}\text{s}^{-1}$
k_b^-	Ca^{2+} -independent rate constant from $O_2 \rightarrow O_1$	385.9	s^{-1}
k_c^+	Ca^{2+} -independent rate constant from $O_1 \rightarrow C_2$	1.75	s^{-1}
k_c^-	Ca^{2+} -independent rate constant from $C_2 \rightarrow O_1$	0.1	s^{-1}
K_m	Half-maximal inhibition of RTC by Mg^{2+}	$200^\dagger$	μM
n_m	Hill coefficient of Mg^{2+} inhibition of RTC	$1.1^\ddagger$	unitless

As reported by O'Brien (29)[†] and Valdivia et al. (28)[‡].

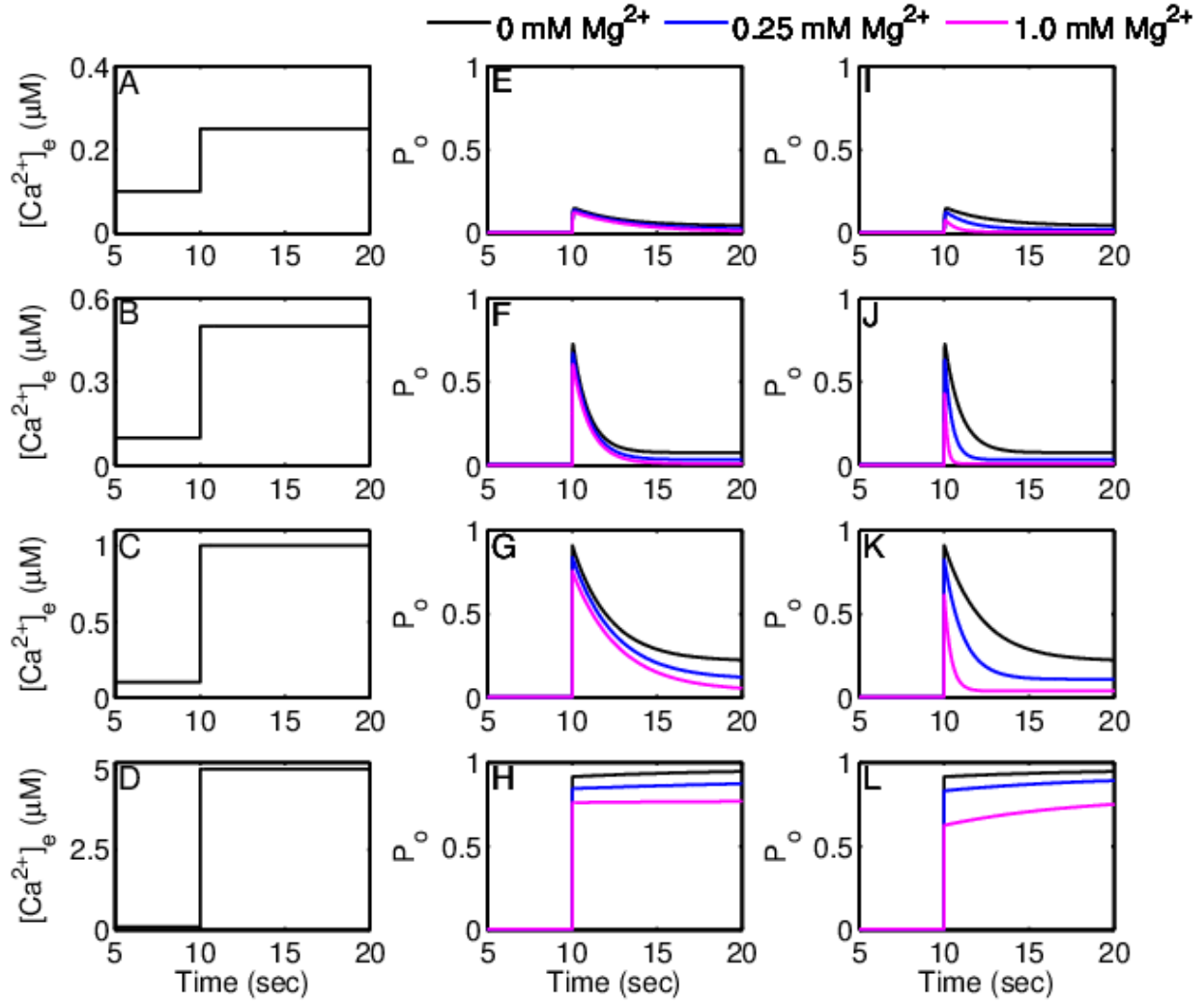


Figure A.3: Model simulations showing the Ca^{2+} dependent RTC adaptation phenomena with different cytosolic Mg^{2+} concentrations. To simulate the phenomena, at time $t = 10$ sec, the cytosolic Ca^{2+} concentration was fixed to $0.25 \mu\text{M}$ (A), $0.5 \mu\text{M}$ (B), $1 \mu\text{M}$ (C), and $5 \mu\text{M}$ (D). (E-H) The effect of Mg^{2+} is accounted for by the Mg^{2+} binding step shown in Figure 1D of the main-text. Simulations of the RTC open probability which is the sum of P_{O1} and P_{O2} with different cytosolic Mg^{2+} concentrations: 0 mM (black), 0.25 mM (blue), and 1 mM (magenta). It is apparent that the effect of cytosolic Mg^{2+} over the Ca^{2+} dependent adaptation of the RTC is not robust, but the RTC open probability is affected. Note that for the range of Ca^{2+} in Boelen's experiments (A) the open probability is minimally affected (E). (I-L) RTC open probability simulations using the alternate model which is obtained by assuming rapid binding of cytosolic Mg^{2+} ions with the O_1 state instead of C_2 state. Mathematically, it is equivalent to multiplying k_c^+ by $f(\text{Mg}^{2+})^{-1}$. It is apparent that this model exhibits the rapid adaptation phenomenon in the presence of cytosolic Mg^{2+} ions similar to cardiac SR-RyR (28). However, this model was not able to reproduce the

Ca^{2+} uptake experiments on isolated cardiac mitochondria (Figure A.4). Also note that, at high cytosolic Ca^{2+} concentrations (H, L), both of the models do not adapt, which is consistent with the original Keizer-Levine model.

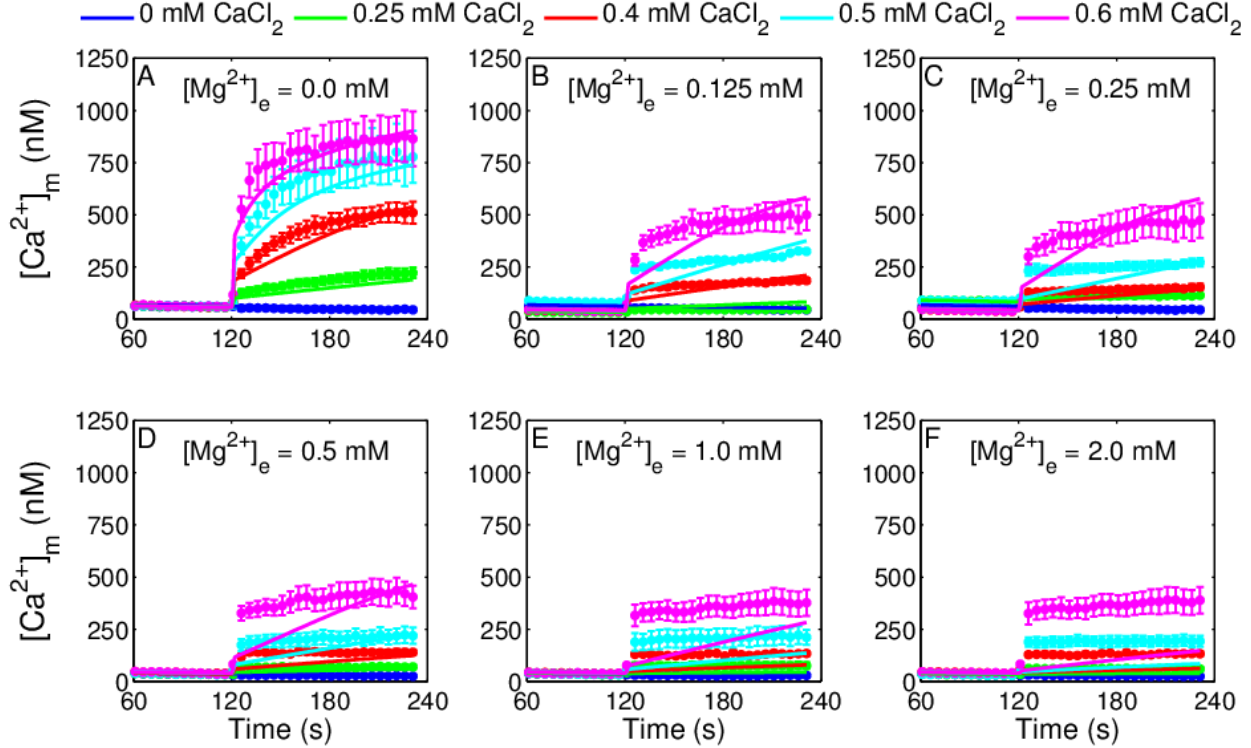


Figure A.4: Model simulations of the Ca^{2+} uptake experiments with RTC model exhibiting fast adaptation in the presence of cytosolic Mg^{2+} ions. Experimentally measured matrix $[\text{Ca}^{2+}]$ dynamics (closed circles with error bars) and model simulations (solid lines) in response to added CaCl_2 boluses of 0 mM (blue), 0.25 mM (green), 0.4 mM (red), 0.5 mM (cyan) and 0.6 mM (magenta). The experimental buffer contained either 0 mM (A), 0.125 mM (B), 0.25 mM (C), 0.5 mM (D), 1 mM (E), or 2 mM (F) MgCl_2 , also containing 1 mM of EGTA. The estimated parameters corresponding to the mitochondrial Ca^{2+} sequestration system, CU and RTC that provide these model simulations are listed in Table 1 of the main document.

Mitochondrial Ca^{2+} Sequestration System

Total Ca^{2+} in the mitochondria matrix is the sum of free Ca^{2+} and bound Ca^{2+} :

$$[\text{TCa}]_m = [\text{Ca}^{2+}]_m + [\text{BCa}]_m, \quad (\text{A.18})$$

Differentiating the above equation w.r.t. time and rearranging for $[\text{Ca}^{2+}]_m$ we get:

$$\frac{d[\text{Ca}^{2+}]_m}{dt} = \frac{1}{\beta_{\text{Ca},m}} \frac{d[\text{TCa}]_m}{dt} \text{ with } \beta_{\text{Ca},m} = 1 + \frac{\partial[\text{BCa}]_m}{\partial[\text{Ca}^{2+}]_m}. \quad (\text{A.19})$$

$\beta_{\text{Ca},m}$ is called the mitochondrial Ca^{2+} buffering capacity which is $\gg 1$ for a highly buffered environment; therefore, under such a highly buffered environment large changes in $[\text{TCa}]_m$ would lead to relatively small changes in $[\text{Ca}^{2+}]_m$. The minimal model of mitochondrial Ca^{2+} buffering capacity used in the integrated mitochondrial model is that initially proposed by Bazil et al. (1) and has the following form:

$$\beta_{\text{Ca},m} = \left(1 + \sum_i \frac{n_i^2 [\text{Ca}^{2+}]_m^{n_i-1} [B_{\text{Ca},i}]_m}{K_{\text{Ca},i}^{n_i} \left(1 + [\text{Ca}^{2+}]_m^{n_i} / K_{\text{Ca},i}^{n_i} \right)^2} \right), \quad (\text{A.20})$$

where i spans over different classes within the buffering model, $[B_{\text{Ca},i}]_m$ is the total concentration of i^{th} class buffer, $K_{\text{Ca},i}$ is the Ca^{2+} affinity of i^{th} class buffer, and n_i is the number of binding sites for i^{th} class buffer. In line with Bazil et al. (1), we used two classes of Ca^{2+} buffer (i.e., Class 1 and Class 2) to explain the mitochondrial Ca^{2+} sequestration system. Associated parameter values are listed in Table 1 of the main document.

Appendix B: Dynamic Mass Balance Equations

The following differential equations constitute the simplified model of mitochondrial electron transport system and oxidative phosphorylation. This model is further augmented by the differential equations for metal ions (K^+ , Mg^{2+} , Ca^{2+} and Na^+) in the matrix and extra-matrix spaces and the differential equation for H^+ in matrix space.

$$\begin{aligned} d\Delta\Psi/dt &= (+4J_{\text{C1}} + 2J_{\text{C3}} + 4J_{\text{C4}} - n_{\text{F1F0}}J_{\text{F1F0}} - J_{\text{ANT}} - J_{\text{Hleak}} - 4J_{\text{CU}} - 2J_{\text{RTC}} \\ &\quad - (n_{\text{NCE}} - 2)J_{\text{NCE}}) / C_{\text{IMM}} \\ d[\text{NADH}]_x/dt &= (+J_{\text{DH}} - J_{\text{C1}}) / W_x \\ d[\text{UQH}_2]_x/dt &= (+J_{\text{C1}} - J_{\text{C3}}) / W_x \\ d[\text{cyt-c}(\text{red})^{2+}]_i/dt &= (+2J_{\text{C3}} - 2J_{\text{C4}}) / W_i \\ d[\text{ATP}]_x/dt &= (+J_{\text{F1F0}} - J_{\text{ANT}}) / W_x \\ d[\text{ADP}]_x/dt &= (-J_{\text{F1F0}} + J_{\text{ANT}}) / W_x \\ d[\text{Pi}]_x/dt &= (-J_{\text{F1F0}} + J_{\text{PHT}}) / W_x \end{aligned} \quad (\text{B.1})$$

$$\begin{aligned}
d[\text{ATP}]_i/dt &= (+J_{\text{ATPt}} + J_{\text{ANT}})/W_i \\
d[\text{ADP}]_i/dt &= (+J_{\text{ADPt}} - J_{\text{ANT}})/W_i \\
d[\text{Pi}]_i/dt &= (+J_{\text{Pit}} - J_{\text{PHT}})/W_i \\
d[\text{ATP}]_e/dt &= -J_{\text{ATPt}}/W_e \\
d[\text{ADP}]_e/dt &= -J_{\text{ADPt}}/W_e \\
d[\text{Pi}]_e/dt &= -J_{\text{Pit}}/W_e.
\end{aligned} \tag{B.2}$$

The following state variables are computed from the total concentration pools based on mass conservations:

$$\begin{aligned}
[\text{NAD}^+]_x &= \text{NAD}_{\text{tot}} - [\text{NADH}]_x \\
[\text{UQ}] &= \text{UQ}_{\text{tot}} - [\text{UQH}_2] \\
[\text{cyt-c(ox)}^{3+}] &= \text{cyt-c}_{\text{tot}} - [\text{cytC(red)}^{2+}] \\
[\text{ADP}]_x &= \text{A}_{\text{tot}} - [\text{ATP}]_x
\end{aligned} \tag{B.3}$$

Appendix C: Differential Equations for Cations: H^+ , K^+ , Mg^{2+} , Ca^{2+} and Na^+

The cation differential equations for the mitochondrial and extra-mitochondrial compartments are derived using the method outlined in Vinnakota et al. (30). The general method will be presented with the understanding that compartment specific concentrations and transport rates are to be used where appropriate. In some cases, compartment specific equations will be given. Due to the large expressions resulting from the derivation, the method used to obtain them is presented versus explicitly showing all the terms that enter the differential equations (each differential equation would take several pages). The generalized system of equations relating the cation differential equations is:

$$\begin{bmatrix} \frac{d[\text{H}^+]}{dt} \\ \frac{d[\text{Na}^+]}{dt} \\ \frac{d[\text{K}^+]}{dt} \\ \frac{d[\text{Mg}^{2+}]}{dt} \\ \frac{d[\text{Ca}^{2+}]}{dt} \end{bmatrix} = \begin{bmatrix} 1 + H_{\partial H} & H_{\partial K} & H_{\partial Na} & H_{\partial Mg} & H_{\partial Ca} \\ Na_{\partial H} & 1 + Na_{\partial Na} & Na_{\partial K} & Na_{\partial Mg} & Na_{\partial Ca} \\ K_{\partial H} & K_{\partial Na} & 1 + K_{\partial K} & K_{\partial Mg} & K_{\partial Ca} \\ Mg_{\partial H} & Mg_{\partial K} & Mg_{\partial Na} & 1 + Mg_{\partial Mg} & Mg_{\partial Ca} \\ Ca_{\partial H} & Ca_{\partial K} & Ca_{\partial Na} & Ca_{\partial Mg} & 1 + Ca_{\partial Ca} \end{bmatrix}^{-1} \begin{bmatrix} \Phi_H \\ \Phi_{Na} \\ \Phi_K \\ \Phi_{Mg} \\ \Phi_{Ca} \end{bmatrix}, \tag{C.1}$$

where $X_{\partial Y}$ is the partial derivative of the concentration of *bound* X with respect to Y (Y can equal X) and Φ_X is the flux of X into or out of the compartment.

Assuming higher order cation binding is negligible, the partial derivative expressions are defined below where N_r is the number of reactants, L_i is the i^{th} ligand, K_i^j is the dissociation constant for the i^{th} ligand and j^{th} cation couple and P_i is the binding polynomial for the i^{th} ligand as originally defined by Alberty (10).

$$H_{\partial H} = \frac{\partial[H_{\text{bound}}]}{\partial[H^+]} = \sum_{i=1}^{N_r} \frac{[L_i] \left(1 + [Na^+] / K_i^{Na} + [K^+] / K_i^K + [Mg^{2+}] / K_i^{Mg} + [Ca^{2+}] / K_i^{Ca}\right)}{K_i^H P_i^2} \quad (\text{C.2})$$

$$H_{\partial K} = \frac{\partial[H_{\text{bound}}]}{\partial[K^+]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^H}{K_i^K P_i^2}$$

$$H_{\partial Na} = \frac{\partial[H_{\text{bound}}]}{\partial[Na^+]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^H}{K_i^{Na} P_i^2}$$

$$H_{\partial Mg} = \frac{\partial[H_{\text{bound}}]}{\partial[Mg^{2+}]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^H}{K_i^{Mg} P_i^2}$$

$$H_{\partial Ca} = \frac{\partial[H_{\text{bound}}]}{\partial[Ca^{2+}]} = -[H^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^H}{K_i^{Ca} P_i^2} \quad (\text{C.3})$$

$$Na_{\partial H} = \frac{\partial[Na_{\text{bound}}]}{\partial[H^+]} = -[Na^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^{Na}}{K_i^H P_i^2}$$

$$Na_{\partial Na} = \frac{\partial[Na_{\text{bound}}]}{\partial[Na^+]} = \sum_{i=1}^{N_r} \frac{[L_i] \left(1 + [H^+] / K_i^H + [K^+] / K_i^K + [Mg^{2+}] / K_i^{Mg} + [Ca^{2+}] / K_i^{Ca}\right)}{K_i^{Na} P_i^2}$$

$$Na_{\partial K} = \frac{\partial[Na_{\text{bound}}]}{\partial[K^+]} = -[Na^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^{Na}}{K_i^K P_i^2}$$

$$Na_{\partial Mg} = \frac{\partial[Na_{\text{bound}}]}{\partial[Mg^{2+}]} = -[Na^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^{Na}}{K_i^{Mg} P_i^2}$$

$$Na_{\partial Ca} = \frac{\partial[Na_{\text{bound}}]}{\partial[Ca^{2+}]} = -[Na^+] \sum_{i=1}^{N_r} \frac{[L_i] / K_i^{Na}}{K_i^{Ca} P_i^2}$$

$$K_{\partial H} = \frac{\partial[K_{bound}]}{\partial[H^+]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^K}{K_i^H P_i^2}$$

$$K_{\partial Na} = \frac{\partial[K_{bound}]}{\partial[Na^+]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^K}{K_i^{Na} P_i^2}$$

$$K_{\partial K} = \frac{\partial[K_{bound}]}{\partial[K^+]} = \sum_{i=1}^{Nr} \frac{[L_i] \left(1 + [H^+] / K_i^H + [Na^+] / K_i^{Na} + [Mg^{2+}] / K_i^{Mg} + [Ca^{2+}] / K_i^{Ca} \right)}{K_i^K P_i^2}$$

$$K_{\partial Mg} = \frac{\partial[K_{bound}]}{\partial[Mg^{2+}]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^K}{K_i^{Mg} P_i^2}$$

$$K_{\partial Ca} = \frac{\partial[K_{bound}]}{\partial[Ca^{2+}]} = -[K^+] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^K}{K_i^{Ca} P_i^2}$$

$$Mg_{\partial H} = \frac{\partial[Mg_{bound}]}{\partial[H^+]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Mg}}{K_i^H P_i^2}$$

$$Mg_{\partial Na} = \frac{\partial[Mg_{bound}]}{\partial[Na^+]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Mg}}{K_i^{Na} P_i^2}$$

$$Mg_{\partial K} = \frac{\partial[Mg_{bound}]}{\partial[K^+]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Mg}}{K_i^K P_i^2}$$

$$Mg_{\partial Mg} = \frac{\partial[Mg_{bound}]}{\partial[Mg^{2+}]} = \sum_{i=1}^{Nr} \frac{[L_i] \left(1 + [H^+] / K_i^H + [Na^+] / K_i^{Na} + [K^+] / K_i^K + [Ca^{2+}] / K_i^{Ca} \right)}{K_i^{Mg} P_i^2} \quad (C.4)$$

$$Mg_{\partial Ca} = \frac{\partial[Mg_{bound}]}{\partial[Ca^{2+}]} = -[Mg^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Mg}}{K_i^{Ca} P_i^2}$$

$$Ca_{\partial H} = \frac{\partial[Ca_{bound}]}{\partial[H^+]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Ca}}{K_i^H P_i^2} \quad (C.5)$$

$$Ca_{\partial Na} = \frac{\partial[Ca_{bound}]}{\partial[Na^+]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Ca}}{K_i^{Na} P_i^2}$$

$$\begin{aligned}
Ca_{\partial K} &= \frac{\partial[Ca_{bound}]}{\partial[K^+]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Ca}}{K_i^K P_i^2} \\
Ca_{\partial Mg} &= \frac{\partial[Ca_{bound}]}{\partial[Mg^{2+}]} = -[Ca^{2+}] \sum_{i=1}^{Nr} \frac{[L_i] / K_i^{Ca}}{K_i^{Mg} P_i^2} \\
Ca_{\partial Ca} &= \frac{\partial[Ca_{bound}]}{\partial[Ca^{2+}]} = \sum_{i=1}^{Nr} \frac{[L_i] \left(1 + [H^+] / K_i^H + [Na^+] / K_i^{Na} + [K^+] / K_i^K + [Mg^{2+}] / K_i^{Mg} \right)}{K_i^{Ca} P_i^2} \quad (C.6)
\end{aligned}$$

$$P_i = 1 + [H^+] / K_i^H + [Na^+] / K_i^{Na} + [K^+] / K_i^K + [Mg^{2+}] / K_i^{Mg} + [Ca^{2+}] / K_i^{Ca}$$

Additional Buffering in the Mitochondrial Compartment

To account for additional buffering not attributed to metabolites and substrates in the mitochondrial compartment, we modified equations (C.2), (C.4), and (C.6) by adding the following terms:

$$\begin{aligned}
H_{\partial H} &= H_{\partial H} + \frac{[B_H]}{K_H \left(1 + [H^+] / K_H \right)^2} \\
Mg_{\partial Mg} &= Mg_{\partial Mg} + \frac{[B_{Mg}]}{K_{Mg} \left(1 + [Mg^{2+}] / K_{Mg} \right)^2} \\
Ca_{\partial Ca} &= Ca_{\partial Ca} + \sum_i \frac{n_i^2 [Ca^{2+}]^{n_i-1} [B_{Ca,i}]}{K_{Ca,i}^{n_i} \left(1 + [Ca^{2+}]^{n_i} / K_{Ca,i}^{n_i} \right)^2} \quad (C.7)
\end{aligned}$$

Note that Equation (C.7) accounts for mitochondrial Ca^{2+} buffering given by Equation (A.20). The parameters associated with it were estimated based on mitochondrial Ca^{2+} uptake experiments of Boelens et al. (5) and Blomeyer et al. (4) and are provided in Table 1 of the main document.

Table C.1. Additional Mitochondrial Buffering Parameters

Parameter	Definition	Value	Reference
$[B_H]$	Total buffer concentration	20 mM	(13)
K_H	H^+ binding constant	$10^{-7.5}$ M	(13)
$[B_{Mg}]$	Total buffer concentration	5.6 mM	(31)
K_{Mg}	Mg^{2+} binding constant	5.9 μ M	(31)

Additional Buffering in the Extra-Mitochondrial Compartment

EGTA were included in the incubation medium for the conditions simulated in the main paper. EGTA buffers H^+ and Ca^{2+} . Moreover, EGTA is primarily dibasic in the pH range simulated. To account for this additional buffer in the extra-matrix compartment equations (C.2), (C.3), (C.4), and (C.5) were modified by adding the following terms:

$$\begin{aligned}
 H_{\partial H} &= H_{\partial H} + \frac{[EGTA](2[H^+] + K_{HEGTA}^H)([Ca^{2+}] + K_{EGTA}^{Ca})}{K_{EGTA}^{Ca} K_{EGTA}^H K_{HEGTA}^H P_{EGTA}^2} \\
 H_{\partial Ca} &= H_{\partial Ca} - \frac{[EGTA][H^+]([H^+] + K_{HEGTA}^H)}{K_{EGTA}^{Ca} K_{EGTA}^H K_{HEGTA}^H P_{EGTA}^2} \\
 Ca_{\partial H} &= Ca_{\partial H} - \frac{[EGTA][Ca^{2+}](2[H^+] + K_{HEGTA}^H)}{K_{EGTA}^{Ca} K_{EGTA}^H K_{HEGTA}^H P_{EGTA}^2} \\
 Ca_{\partial Ca} &= Ca_{\partial Ca} + \frac{[EGTA]([H^+]^2 + [H^+]K_{HEGTA}^H + K_{EGTA}^H K_{HEGTA}^H)}{K_{EGTA}^{Ca} K_{EGTA}^H K_{HEGTA}^H P_{EGTA}^2}
 \end{aligned}$$

where the binding polynomial for EGTA is:

$$P_{EGTA} = 1 + [H^+] / K_{EGTA}^H + [H^+]^2 / K_{HEGTA}^H / K_{EGTA}^H + [Ca^{2+}] / K_{EGTA}^{Ca}$$

Flux Terms

The terms representing flux into a given compartment are defined as:

$$\begin{aligned}
 \Phi_H &= -\sum_{i=1}^{Nr} \frac{\partial[H_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + \sum_{k=1}^{Nk} n_k J_k + J_t^H \\
 \Phi_{Na} &= -\sum_{i=1}^{Nr} \frac{\partial[Na_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^{Na} \\
 \Phi_K &= -\sum_{i=1}^{Nr} \frac{\partial[K_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^K \\
 \Phi_{Mg} &= -\sum_{i=1}^{Nr} \frac{\partial[Mg_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^{Mg}
 \end{aligned}$$

$$\Phi_{Ca} = -\sum_{i=1}^{Nr} \frac{\partial[Ca_{bound}]}{\partial[L_i]} \frac{d[L_i]}{dt} + J_t^{Ca}$$

where N_k is the number of reactions, n_k is the stoichiometric coefficient of k^{th} reaction, J_k is the k^{th} reaction rate and J_t^i is the i^{th} cation transport rate into the compartment. The ligand dependent partial derivative expressions are defined as:

$$\frac{\partial[H_{bound}]}{\partial[L_i]} = [H^+] \sum_{i=1}^{Nr} \frac{1}{K_i^H P_i}$$

$$\frac{\partial[Na_{bound}]}{\partial[L_i]} = [Na^+] \sum_{i=1}^{Nr} \frac{1}{K_i^{Na} P_i}$$

$$\frac{\partial[K_{bound}]}{\partial[L_i]} = [K^+] \sum_{i=1}^{Nr} \frac{1}{K_i^K P_i}$$

$$\frac{\partial[Mg_{bound}]}{\partial[L_i]} = [Mg^{2+}] \sum_{i=1}^{Nr} \frac{1}{K_i^{Mg} P_i}$$

$$\frac{\partial[Ca_{bound}]}{\partial[L_i]} = [Ca^{2+}] \sum_{i=1}^{Nr} \frac{1}{K_i^{Ca} P_i}$$

Compartment Reactions and Fluxes

The generation of protons by biochemical reactions in mitochondrial is defined as

$$\sum_{k=1}^{Nk} n_k J_k = J_{DH}$$

This term is zero in the extra-mitochondrial compartment. The transport of cations into the mitochondrial compartment is defined as

$$J_t^H = \left((nH_{F_1F_0} - 1) J_{F_1F_0} + 2J_{PIC} + J_{KHE} + J_{NHE} + 2J_{CHE} + J_{Hleak} - 5J_{C1} - 2J_{C3} - 4J_{C4} \right) \quad (C.8)$$

$$J_t^{Na} = (3J_{NCE} - J_{NHE}) \quad (C.9)$$

$$J_t^K = -J_{KHE}; J_t^{Mg} = 0 \quad (C.10)$$

$$J_t^{Ca} = (2J_{CU} + J_{RTC} - J_{NCE} - J_{CHE}) \quad (C.11)$$

The transport terms for the extra-mitochondrial compartment are negative of their mitochondrial compartment.

Appendix D: List of reactants, abbreviation, reference species, Gibbs free energy of formation, and binding constants (298.15 K, 1 M reactants, I = 0.17 M, P = 1 atm). All pK values are taken from NIST database (pK^a: temperature T = 298 K and ionic strength I = 0.1) and further corrected (pK^b: T = 310.15 K and I = 0.17) for the new temperature and ionic strength based on the following algorithm:

Step 1: Calculate ΔH (I = 0, T = T_{old}) from ΔH (I = I_{old}, T = T_{old}):

$$\Delta H(I = 0) = \Delta H_{I=I_{old}} - RT^2 \cdot (\partial \alpha / \partial T) \cdot \left(\sqrt{I} / (1 + B\sqrt{I}) \right) \cdot \sum_{i=1}^N v_i^2 z_i^2 \quad (D.1)$$

where $RT^2 \left(\frac{\partial \alpha}{\partial T} \right) = 1.4775$, B = 1.6, R = 8.314 × 10⁻³, I = ionic strength.

Step 2: Calculate K (I = 0, T = T_{old}) from K (I = I_{old}, T = T_{old}):

$$K(I = 0) = K_{I=I_{old}} \prod_{i=1}^N \gamma_i^{-v_i}; T = T_{old}, \quad (D.2)$$

$$\gamma = \exp \left(\frac{-Az_i^2 \sqrt{I}}{1 + B\sqrt{I}} \right), B = 1.6, A = 1.10708 - (1.54508 \times 10^{-3})T + (5.95584 \times 10^{-6})T^2,$$

Step 3: Calculate K (I = 0, T = T_{new}) from K (I = 0, T = T_{old}):

$$K_{I=I_{new}} = K_{I=0} \cdot \Delta H_{I=0} \cdot \left(\frac{T_{new} - T_{old}}{2.303T_{new}T_{old}} \right) \quad (D.3)$$

Step 4: Calculate K (I_{new}, T = T_{new}) from K (I = 0, T = T_{new}):

$$K_{new} = K_{I=I_{new}} \cdot \prod_{i=1}^N \gamma_i^{-v_i} \quad (D.4)$$

Reactant	Reference species	$\Delta_f G_o$ (kJ/mol)	Metal ion-bound species	pK ^a	pK ^b
ATP	ATP ⁴⁻	-2771.00	HATP ³⁻	6.48	6.64
			H ₂ ATP ²⁻	3.99	3.77
			KATP ³⁻	1.17	1.02
			MgATP ²⁻	4.19	3.98
			MgHATP ⁻	2.32	2.30
			Mg ₂ ATP	1.70	1.67
			CaATP ²⁻	3.86	3.62
			CaHATP ⁻	2.16	1.97
			NaATP ³⁻	1.17	1.12
ADP	ADP ³⁻	-1903.96	HADP ²⁻	6.38	6.66
			H ₂ ADP ⁻	3.87	3.74
			KADP ²⁻	1.00	1.77
			MgADP ⁻	3.23	3.09
			MgHADP	1.59	3.48
			Mg ₂ ADP ⁺	1.0	1.48
			CaADP ⁻	2.86	1.7
			CaHADP	1.48	2.69
			NaADP ³⁻	1.00	1.36
			NaHADP ²⁻	0.01	1.01
Pi	HPO ₄ ²⁻	-1098.27	H ₂ PO ₄ ⁻	6.70	6.59
			KHPO ₄ ⁻	0.50	0.42
			KH ₂ PO ₄	0.20	0.16
			K ₂ HPO ₄	0.50	0.17
			MgH PO ₄	1.81	1.73
			Mg H ₂ PO ₄ ⁺	1.19	1.10
			Ca H PO ₄	1.60	1.52
			CaH ₂ PO ₄ ⁺	0.60	0.52
			Na HPO ₄ ⁻	0.20	0.21
			NaH ₂ PO ₄	0.50	0.49
EGTA	EGTA ⁴⁻		H EGTA ³⁻	9.40	9.08
			H ₂ EGTA ²⁻	80.79	8.52
			KEGTA ³⁻	1.22	1.06
					5.12

			MgEGTA ²⁻	5.28	7.50
			MgHEGTA ⁻	7.59	10.32
			CaEGTA ²⁻	10.86	3.72
			CaHEGTA ⁻	3.81	1.19
			Na EGTA ³⁻	1.22	0.01
			NaH EGTA ²⁻	0.01	

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